

Use Case Specification: UC-01

Purchase Item & Lock Escrow

1. Use Case Information

Use Case Name: Purchase Item and Lock Escrow

Primary Actor: Buyer

Secondary Actor: Payment Gateway

Brief Description: This use case describes the process where a buyer purchases a second-hand item, resulting in the system securely holding the payment in escrow and triggering the inspection process.

2. Preconditions

- The Buyer is authenticated and logged into the marketplace system.
- The selected second-hand product is actively listed and available for purchase.
- The system has a stable connection to the external Payment Gateway.

3. Postconditions

- The Buyer's payment is securely locked in the system's escrow account.
- The product's order state is successfully transitioned to 'Under Inspection'.
- A notification is dispatched to the Inspection Agent to commence physical verification.

4. Main Success Scenario (Basic Flow)

1. The Buyer selects a desired second-hand product and clicks the "Purchase" button.
2. The System calculates the total amount (including any escrow/platform fees) and displays the payment interface.
3. The Buyer inputs their payment details and confirms the transaction.
4. The System validates the input format and forwards the transaction request to the Payment Gateway.
5. The Payment Gateway processes the request and sends an authorization success response to the System.
6. The System securely routes and locks the funds into the escrow holding account.
7. The System updates the order state of the item to 'Under Inspection'.
8. The System automatically notifies the assigned Inspection Agent that a new item requires physical verification.
9. The System displays a transaction success message and digital receipt to the Buyer.

5. Alternate Flow

5a. Payment Authorization Failed (Branches from Step 5)

- The Payment Gateway declines the transaction (e.g., insufficient funds, invalid card) and sends a failure response.
- The System halts the escrow lock and displays a "Payment Failed" error message to the Buyer.
- The System prompts the Buyer to either update their payment details, select an alternative payment method, or cancel the transaction.
- If the Buyer successfully provides a new payment method, the flow resumes at Step 4.
- If the Buyer cancels the transaction or fails three consecutive payment attempts, the System terminates the use case and the item remains actively listed on the marketplace.